

Lecture 9 – Performance Evaluation

INT104 Artificial Intelligence

Performance Evaluation

Dr. Shengchen Li

Associate Professor

Department of Intelligence Science

Xi'an Jiaotong–Liverpool University

Evaluating Classification Methods

- Performance (for Binary classification)
 - classifier performance: predicting class label compared with groundtruth
 - True Positive (TP), True Negative (TN), False Positive (FP), False Negative (FN)
- Complexity
 - Time to construct the model (training time)
 - ◆ must be tolerable
 - ◆ can be large (lengthy)
 - Time to use the model (inference time)
 - ◆ must be tolerable
 - ◆ need for good data structures
 - Operations to train / use the model
 - Number of parameters in the models
- Robustness
 - handling noise and missing values
 - handling incorrect training data

Confusion Matrix

- TP, FP, FN and TN form a matrix named confusion matrix. There are many parameters that can be calculated according to the confusion matrix, which evaluate the resulting system from different perspectives.

		True condition			
		Condition positive	Condition negative	Prevalence $= \frac{\sum \text{Condition positive}}{\sum \text{Total population}}$	Accuracy (ACC) = $\frac{\sum \text{True positive} + \sum \text{True negative}}{\sum \text{Total population}}$
Predicted condition	Predicted condition positive	True positive	False positive, Type I error	Positive predictive value (PPV), Precision = $\frac{\sum \text{True positive}}{\sum \text{Predicted condition positive}}$	False discovery rate (FDR) = $\frac{\sum \text{False positive}}{\sum \text{Predicted condition positive}}$
	Predicted condition negative	False negative, Type II error	True negative	False omission rate (FOR) = $\frac{\sum \text{False negative}}{\sum \text{Predicted condition negative}}$	Negative predictive value (NPV) = $\frac{\sum \text{True negative}}{\sum \text{Predicted condition negative}}$
		True positive rate (TPR), Recall, Sensitivity, probability of detection, Power = $\frac{\sum \text{True positive}}{\sum \text{Condition positive}}$	False positive rate (FPR), Fall-out, probability of false alarm $= \frac{\sum \text{False positive}}{\sum \text{Condition negative}}$	Positive likelihood ratio (LR+) $= \frac{\text{TPR}}{\text{FPR}}$	Diagnostic odds ratio (DOR) = $\frac{\text{LR+}}{\text{LR-}}$ F ₁ score = $2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$
		False negative rate (FNR), Miss rate $= \frac{\sum \text{False negative}}{\sum \text{Condition positive}}$	Specificity (SPC), Selectivity, True negative rate (TNR) $= \frac{\sum \text{True negative}}{\sum \text{Condition negative}}$	Negative likelihood ratio (LR-) $= \frac{\text{FNR}}{\text{TNR}}$	

Confusion Matrix

Metric	Full Name	Definition
Recall (Sensitivity)	True Positive Rate	(1)
Fall-out	False Positive Rate	(3)
Miss rate	False Negative Rate	(2)
Specificity (Selectivity)	True Negative Rate	(4)

$$\text{recall} = \frac{TP}{TP + FN} = \frac{TP}{n^+} \quad (1)$$

$$\text{fallout} = \frac{FP}{TN + FP} = \frac{FP}{n^-} \quad (3)$$

$$\text{miss} = \frac{FN}{TP + FN} = \frac{FN}{n^+} \quad (2)$$

$$\text{specificity} = \frac{TN}{TN + FP} = \frac{TN}{n^-} \quad (4)$$

NB: Recall is also named sensitivity

Confusion Matrix

More general measurements:

$$\text{precision} = \frac{TP}{TP + FP} \quad (5)$$

$$accuracy = \frac{TN + TP}{TP + TN + FP + FN} = \frac{TN + TP}{n} \quad (6)$$

A good measurement should balance the multiple aspects such as F_1 Score (F-Measure)

$$F_1score = \frac{2 \cdot TP}{2 \cdot TP + FP + FN} = 2 \cdot \frac{precision \cdot recall}{precision + recall} \quad (7)$$

Exercise

Please calculate Accuracy, Specificity, Recall and F_1 score for the givencases

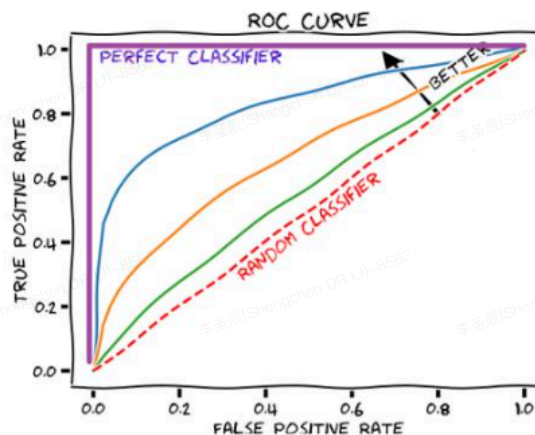
TP	FP	FN	TN	Accu	Recall	Precision	F1 Sc
63	28	37	72				
77	77	23	23				
24	88	76	12				
76	12	24	88				

Receiver Operation Characteristics (ROC)

Assume that the following scores are obtained from classifiers and you would like to set a threshold for decision making. How do you evaluate the system performance of different threshold values?

Predict Score	Label
0.11	1
0.35	0
0.92	0
0.51	1
0.49	0
0.81	1
0.26	0
0.63	0
0.72	1
0.25	1

Receiver Operation Characteristics (ROC)



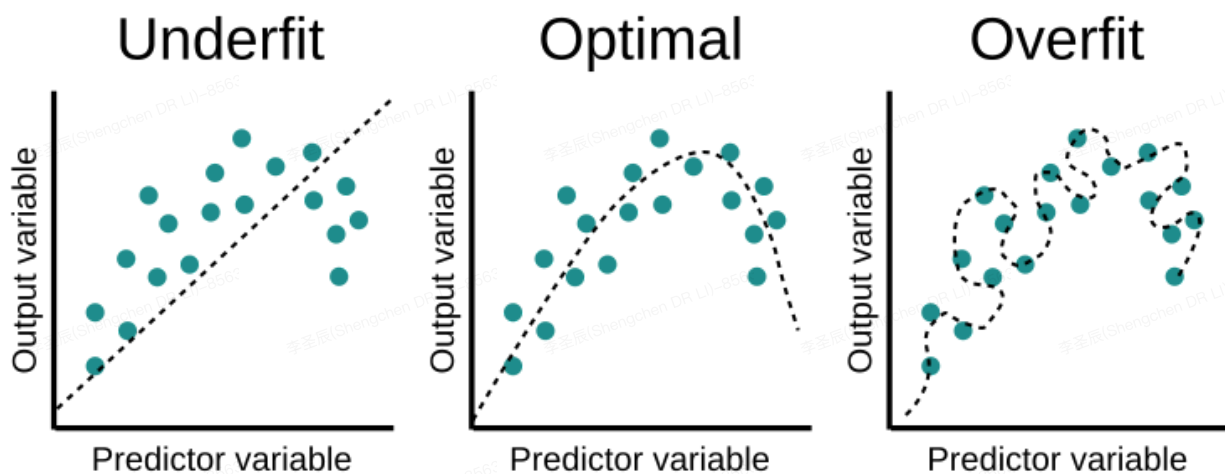
- One of the measurement for ROC test is Area under Curve (AUC)
- For $AUC > 0.5$, the system performance is better than random guess

Cross Validation

- Ideally, the training data should have the same distribution in practice, which never happens ...
- How do we test the performance of resulting models?
- We could use a part of training data as the validation set
- By performing cross validation, we expect the validation set must be a good representative for the whole data

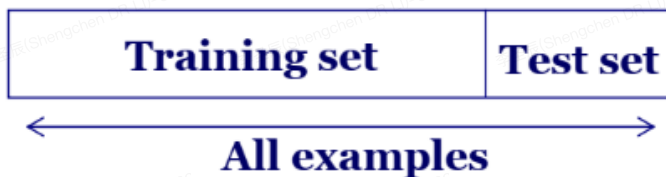
Overfit

- One of the most common objective for cross validation is to avoid overfitting
- If a system is too complex, the performance of the system could be too good to be true



Data Partitioning Methods for X-Validation

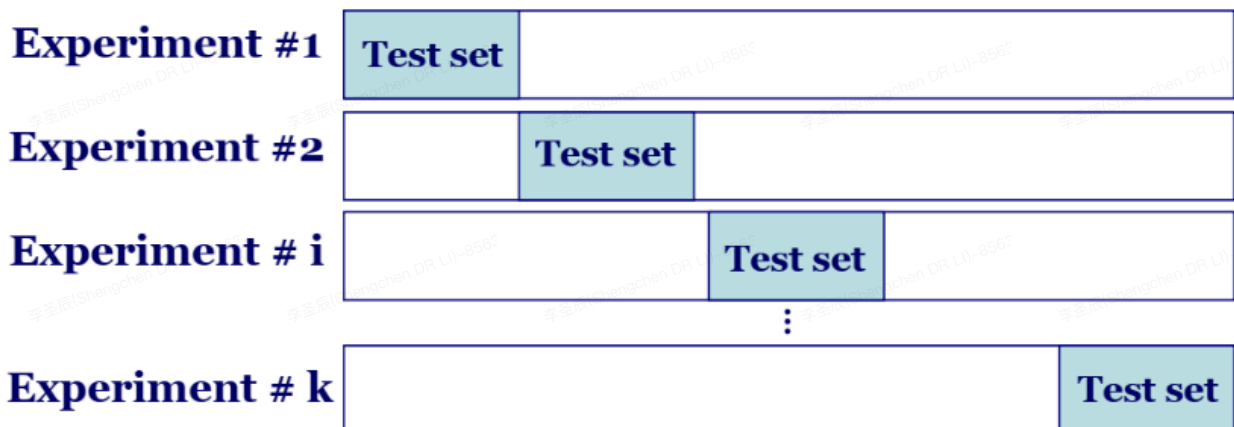
- Random Sampling
 - data is randomly partitioned into two separate sets
 - Assumption: data is uniformly distributed



- Bootstrap
 - resample with replacement n sample of original data as training set.
 - some numbers in the original sample may be included several times in the bootstrap sample
 - can be used to estimate data distribution

k-fold Cross Validation

- k-fold (k is usually set to 5, 10 and other numbers)
 - Randomly partition the data into k mutually exclusive subsets, each approximately equal size
 - At i -th iteration, use D_i as test set and others as training set
 - The mean of measures obtained in iterations used as the output of performance measure



Leave-one-out Cross Validation

- k folds where k is the number of samples.
- The method is particularly useful for small sized dataset.

Three-way Data Splits

- If model selection and true error estimates are to be computed simultaneously, the data needs to be divided into three disjoint sets
 - Training set: a set of examples used for learning: to fit the parameters of the classifier
 - Validation set: a set of examples used to tune the parameters of a classifier
 - Test set: a set of examples used only to assess the performance of a fully-trained classifier
- Why separate test and validation sets?
 - The error rate estimate of the final model on validation data will be biased (smaller than the true error rate) since the validation set is used to select the final model
 - After assessing the final model with the test set, YOU MUST NOT tune the model any further

Model Selection Criteria

- For model-based method, model likelihood could be used as a metric of system performance
- Please remember:
 - the model likelihood can only be compared with the same testing dataset
 - the model likelihood should be compared in the basis of per sample
- To prevent the overfitting problem, model complexity should be considered



The typical way to evaluate a model is model selection criteria.

Akaike's Information Criterion (AIC) and Bayesian Information Criterion (BIC) are

$$\text{AIC} = 2k - 2 \ln(\mathcal{L})$$

$$\text{BIC} = k \ln(n) - 2 \ln(\mathcal{L})$$